Lifestyle Factors Show More Consistent Associations with Gamma-Glutamyl Transferase Than Alanine Aminotransferase: Evidence from the 2011–12 Australian Health Survey

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Abstract

Alanine aminotransferase (ALT) and gamma-glutamyl transferase (GGT) are liver biomarkers potentially associated with lifestyle behaviours. However, evidence concerning alcohol risk, smoking duration and physical activity remains limited in Australian populations. Using cross-sectional data from the 2011–12 Australian Health Survey, we examined these associations in adults using multinomial logistic regression of Low, Borderline and High biomarker categories, adjusting for demographic, anthropometric and socioeconomic factors. Complete-case samples included 4,766 participants for alcohol and smoking analysis and 4,762 for physical activity. High-risk versus low-risk drinking was associated with increased adjusted relative odds of High versus Low GGT (odds ratio 3.66, 95% confidence interval 2.57–5.22), with evidence of an alcohol-risk gradient. Longer smoking duration also showed a positive trend for High GGT ($p = 0.023$), although individual categories varied. ALT associations were weaker and less consistent. Not meeting exercise guidelines was associated with increased adjusted relative odds of High versus Low GGT (1.35, 1.12–1.62), but not High ALT. Meeting guidelines corresponded to a 4.5-percentage-point higher adjusted probability of Low GGT and a 4.0-point lower probability of High GGT. Overall, GGT showed more consistent associations, suggesting potential value for future liver-health assessment and targeted prevention strategies, although causality cannot be inferred.

Introduction

According to the 2019 Global Burden of Disease study, the leading risk factors associated with disease are alcohol, tobacco and psychoactive drug use [1]. The World Health Organisation estimated that 7% of the global population (aged 15+) lived with alcohol use disorder and that alcohol accounted for 4.7% of global deaths, with low-income European and African countries being disproportionately represented [1]. Alcohol dependence and its associated morbidities and mortalities [1–3] continue to also grow in the lower socioeconomic groups of Western countries [3]. Smoking is another modifiable risk factor commonly associated with alcohol dependence and lower socioeconomic status (SES) [2, 4, 5].

More specifically, the incidence of chronic liver diseases, tightly linked with alcohol use and smoking [1, 6], has increased over time and burdened the healthcare system [7]. Habitual alcohol consumption dose- and duration-dependently increases the risk of liver disease, alcoholic cirrhosis, and liver cancer, through its promotion of oxidative stress and inflammation, abnormal lipid profiles, hepatotoxicity, and overall, liver dysfunction [1, 4, 6, 8, 9]. Similarly, cigarette toxins may alter enzymatic and inflammatory pathways in the liver, inducing greater oxidative stress and inflammation [8–10]. However, most liver diseases initially develop without serious signs and symptoms and often present themselves at later stages via jaundice, ascites, and various other features [7]. Questionnaires also usually elicit respondent dishonesty regarding alcohol abuse [11]. These factors motivate objective measures for early detection of alcohol dependence and post-treatment monitoring, particularly in lower SES groups [2–4, 12, 13].

Therefore, haematological biomarkers used in liver function tests, alanine aminotransferase (ALT) and gamma-glutamyl transferase (GGT), can be used as non-invasive means to indicate the extent of liver injury and oxidative stress [2, 14], which is often higher in high-risk alcohol consumers and smokers [2, 8, 12]. ALT is mainly found in the liver and is responsible for the processing of amino acids for energy [7]. Leakage of ALT into the bloodstream thus provides a highly specific measure of hepatocellular damage, where slight elevations may indicate liver disorders in asymptomatic patients [15, 16]. GGT is located in the liver, pancreas, kidneys, and various other locations, and is responsible for degrading extracellular glutathione (GSH) and providing the rate-limiting cysteine required for intracellular de-novo synthesis of GSH [17, 18]. Hence, during heightened oxidative stress, GSH consumption increases, leading to an increased activity of GGT [7, 17, 18]. Though GGT has a lower specificity compared to ALT, it has been shown to have high sensitivity to the oxidative stress induced by cigarette toxins and acetaldehyde (from chronic alcohol abuse), and is also associated with liver inflammation and fatty liver [6–9, 11, 16, 19].

Consequently, favourable lifestyle interventions, such as regular exercise, that aim to lower the number of risk factors linked with liver disease, are important public health and personalised medicine concerns [4, 20]. Indeed, exercise can act as a protective mechanism for liver health by reducing systemic inflammation through modulation of inflammatory and anti-inflammatory cytokines/molecules, reducing fat mass and hepatic steatosis, promoting beta-oxidation and glucose oxidation, and improving insulin sensitivity [16, 20–22]. Thus, exercise has been shown to reduce liver biomarkers ALT [21] and GGT [16] and overall, dose-dependently reduce the risk of liver disease [23], presenting itself as an

important lifestyle intervention for high-risk alcohol consumers and smokers and the broader population.

However, studies report both positive [2,3,24] and null [6,10] associations between alcohol consumption and ALT. Exercise effects on liver biomarkers are also controversial where either no associations [16] or negative associations [20,25] exist. Furthermore, evidence has only found a positive relationship between smoking and GGT, where the lack of association with ALT is thought to be due to the inducement of hepatic oxidative stress [8,9,12] and not direct hepatocellular damage [10,24]. Most of this knowledge originates from non-Australian populations [2–4,6,8–11,14–16,19,22–24], and research on the independent effects of alcohol consumption, smoking, and exercise on ALT and GGT is generally still insufficient and heterogenous [21]. With more clarification of biomarker behaviour in response to lifestyle factors, interventions aimed at promoting healthier habits, educating the vulnerable low SES groups, and emphasising prevention-messaging can be implemented [4].

This study examined whether previously reported associations were replicated in an Australian population. Using the cross-sectional 2011–12 Australian National Health Survey, we examined associations of alcohol consumption, smoking duration and exercise with ALT and GGT. It is hypothesised that (1) ALT and GGT will dose-dependently increase with alcohol consumption risk level, (2) GGT will only increase with smoking, (3) meeting exercise guidelines will be associated with a protective reduction of GGT and ALT and (4) alcohol consumption and smoking may decrease the benefits of exercise on liver health. These findings will contribute to understanding how liver biomarker behaviour can be used as a tool to identify alcohol/smoking-dependence before the onset of liver complications, monitor post-treatment effects in heavy drinkers [3,12,19], and overall improve the effectiveness of early prevention-messaging and lifestyle interventions in high-risk populations [4].

Materials and Methods

Data Source and Study Population

The main data source for the analyses is the 2011–12 National Health Survey component of the Australian Health Survey [30]. The survey contains data on the individuals being surveyed, their lifestyles, households, and relevant biomarkers. The NHS data sets used were `AHSnhs11bbi`, `AHSnhs11bsp`, and `AHSnhs11bhh`, and were linked by person and household record identifiers `ABSPID` and `ABSLID`. Only adults were considered for the analyses.

Variable Preparation and Definitions

The variables used and their types are summarised in Table 1. In general, raw special/magic missing-value codes were recoded to `NA`, and categorical codes were relabelled and converted to factors. `Age` was originally categorical, but was transformed to a numeric variable by taking the midpoint of the binned categories. Medium and high risk alcohol consumers are collapsed into one category in `AlcoholRisk`.

The main response variables were `ALT3` and `GGT3`, which were transformations of the original biomarker variables `ALTRESB` and `GGTRESB` via three-category mappings defined in (1) and (2). These classifications extended the binary categorisation to distinguish borderline from high biomarker levels, informed by published literature on liver enzymes and clinical reference intervals [16, 28, 29].

$$\text{ALT3} = \begin{cases} \text{Low}, & \text{if ALT} \begin{cases} < 30, & \text{if Male} \\ < 25, & \text{if Female} \end{cases} \\ \text{Borderline}, & \text{if ALT} \begin{cases} \in [30, 40], & \text{if Male} \\ \in [25, 30], & \text{if Female} \end{cases} \\ \text{High}, & \text{if ALT} \begin{cases} > 40, & \text{if Male} \\ > 30, & \text{if Female} \end{cases} \end{cases} \tag{1}$$

$$\text{GGT3} = \begin{cases} \text{Low}, & \text{if GGT} \begin{cases} < 45, & \text{if Male} \\ < 30, & \text{if Female} \end{cases} \\ \text{Borderline}, & \text{if GGT} \begin{cases} \in [45, 50], & \text{if Male} \\ \in [30, 35], & \text{if Female} \end{cases} \\ \text{High}, & \text{if GGT} \begin{cases} > 50, & \text{if Male} \\ > 35, & \text{if Female} \end{cases} \end{cases} \tag{2}$$

Table 1. Definitions and coding of variables used in the statistical analyses of the 2011–12 National Health Survey component of the Australian Health Survey (AHS) [30].

Variable	Meaning	Type / coding
`ALT3`	Alanine aminotransferase (ALT) category	Ordinal factor: Low / Borderline / High
`GGT3`	Gamma-glutamyl transferase (GGT) category	Ordinal factor: Low / Borderline / High
`ExerciseGuidelines`	Whether recommended physical-activity guidelines were met	Binary factor: Yes / No
`Age`	Age in years	Continuous numeric
`Sex`	Sex	Binary factor: Female / Male
`Waist`	Waist circumference in cm	Continuous numeric
`AlcoholRisk`	Alcohol consumption risk level	Categorical factor: Never drinker / Former drinker / Low risk / Medium risk / High risk (Analysis 1); Binary factor: Low risk / Medium–High risk (Analysis 2)
`SmokingDuration`	Duration of smoking history	Ordinal factor, grouped by years smoked
`NonSchoolQuals`	Whether a non-school educational qualification was completed	Binary factor: Yes / No
`IncomeDecile`	Income decile	Ordinal factor: deciles 1–10
`DailySmokers`	Number of daily smokers in the household	Ordinal factor: None, One, Two, Three or more

Only complete cases were considered for the analyses, resulting in a sample size of $N = 4766$ for Analysis 1, and $N = 4762$ for Analysis 2. The table of variables used is given below; note that `DailySmokers` is only used in our first analysis.

Statistical Analysis

ALT and GGT were classified as Low, Borderline, or High using sex-specific cut-offs. Ordinal logistic regression was initially considered, but the proportional-odds assumption was violated (Brant test: ALT $p < 0.001$; GGT $p = 0.02$). Multinomial logistic regression (`nnet::multinom`) was therefore used, with Low as the reference outcome. Associations were reported as odds ratios (ORs) with Wald 95% confidence intervals.

Separate ALT and GGT models were fitted using complete cases. For Analysis 1, Model 1 included alcohol risk and smoking duration, with low-risk drinking and never having smoked daily as reference categories. Model 2 additionally adjusted for sex, age, waist circumference, income decile, non-school qualifications, and household daily smokers. For Analysis 2, Model 1 included exercise-guideline status (reference: meeting guidelines), while Model 2 additionally adjusted for the same demographic, anthropometric, and socioeconomic factors, plus alcohol risk and smoking duration. Covariate selection for Analysis 2 was informed by a directed acyclic graph and subject-matter considerations.

For Analysis 1, predictor dependence and multicollinearity were assessed before fitting, with no substantial multicollinearity found. Dose–response relationships were examined using linear trend tests based on orthogonal polynomial contrasts for smoking duration (all participants) and ordered alcohol risk (low, medium, high; current drinkers only). Alcohol-by-smoking interactions were examined using transformed linear models of biomarker bin-midpoint values because of sparse multinomial categories.

Adjusted category probabilities were estimated from Model 2 for both analyses, with continuous covariates held at their means and categorical covariates averaged over their levels. For Analysis 2, probability differences (Exercise Yes minus No) were calculated with 95% confidence intervals, applying Holm adjustment to multiple category-specific comparisons. Interactions between exercise and smoking duration or alcohol risk were explored through subgroup-specific probability differences. Multiple imputation by chained equations (MICE) [31] and generalised additive models were considered as supplementary checks for missing-data sensitivity and potential nonlinearity, respectively.

Results

The population characteristics in Table 2 are based on 4,762 adults from the 2011–12 National Health Survey [30], corresponding to the complete-case sample used for Analysis 2. Analysis 1 used a separately prepared analytic dataset with 4,766 observations, as described in the Methods. Descriptive characteristics include the distributions of ALT and GGT categories, physical activity, alcohol risk, smoking duration, socioeconomic indicators, age, sex, waist circumference, and household daily smokers.

Table 2. Characteristics of the Analysis 2 complete-case sample ($N = 4762$) from the 2011–12 National Health Survey, Australian Health Survey [30]. Numerical variables are reported as mean (IQR), with the IQR displayed as Q_1–Q_3; categorical variables are reported as count and proportion.

Variable	Mean (IQR)		
`Age` (Years)	51.8 (37.0–62.0)		
`Waist` (cm)	94.4 (83.0–104.0)		
	Category	**Count**	**Proportion (%)**
`ExerciseGuidelines`	Yes	2464	51.7
	No	2298	48.3
`ALT3`	Low	3658	76.8
	Borderline	525	11.0
	High	579	12.2
`GGT3`	Low	3870	81.3
	Borderline	268	5.6
	High	624	13.1
`AlcoholRisk`	Low risk	4155	87.3
	Medium/High risk	607	12.7
`Sex`	Female	2519	52.9
	Male	2243	47.1
`NonSchoolQuals`	Yes	3081	64.7
	No	1681	35.3
`SmokingDuration`	Never smoked daily	2538	53.3
	Less than 10 years	512	10.8
	10–19 years	556	11.7
	20–29 years	479	10.1
	30–39 years	352	7.4
	40–49 years	238	5.0
	50 years and over	87	1.8
`DailySmokers`	None	3867	81.2
	One	701	14.7
	Two	181	3.8
	Three or More	13	0.3
`IncomeDecile`	1	301	6.3
	2	473	9.9
	3	539	11.3
	4	560	11.8
	5	448	9.4
	6	448	9.4
	7	478	10.0
	8	482	10.1
	9	546	11.5
	10	487	10.2

Results are then reported for the two primary analyses: associations of alcohol risk and smoking duration with liver biomarker status, followed by associations of physical-activity guideline status with ALT and GGT.

Analysis 1 Results

Alcohol risk, smoking duration and liver biomarker status. Associations were generally more consistent for GGT than ALT across crude and adjusted models (Figure 1).

Alcohol risk. In Model 1, medium-risk and high-risk drinking were associated with higher relative odds of High versus Low GGT compared with low-risk drinking (OR = 1.86, 95% CI: 1.40–2.47, $p < 0.0001$ and OR = 2.93, 95% CI: 2.09–4.13, $p < 0.0001$, respectively). These associations persisted after adjustment (Model 2), strengthening for high-risk drinking (OR = 3.66, 95% CI: 2.57–5.22, $p < 0.0001$; medium risk OR = 1.88, 95% CI: 1.40–2.53, $p < 0.0001$). A similar gradient was observed for Borderline GGT (medium risk OR = 1.73, 95% CI: 1.15–2.59, $p = 0.0084$; high risk OR = 2.21, 95% CI: 1.30–3.74, $p = 0.0032$). In contrast, the adjusted association with High ALT was significant for medium-risk drinking (OR = 1.55, 95% CI: 1.13–2.13, $p = 0.0069$) but not high-risk drinking (OR = 1.37, 95% CI: 0.90–2.09, $p = 0.1415$). Never drinkers and former drinkers were retained in the analytic sample and fitted models but were not examined in detail, as the analysis primarily focused on current drinkers. These groups did not differ significantly from low-risk drinkers in High GGT or High ALT after adjustment. Among former drinkers who last drank 12 or more months ago, the crude association with High GGT (OR = 1.63, 95% CI: 1.19–2.23) was attenuated after adjustment (OR = 1.25, 95% CI: 0.90–1.74).

Smoking duration. In Model 1, longer smoking duration was associated with higher odds of High GGT, reaching OR = 2.35 at both 40–49 years (95% CI: 1.68–3.29) and 50 years or more (95% CI: 1.37–4.05). After adjustment, associations were attenuated but remained significant at 20–29 years (OR = 1.48, 95% CI: 1.10–1.98, $p = 0.0091$), 30–39 years (OR = 1.49, 95% CI: 1.07–2.07, $p = 0.0178$) and 40–49 years (OR = 1.76, 95% CI: 1.20–2.57, $p = 0.0039$), but not at 50 years or more (OR = 1.55, 95% CI: 0.86–2.79, $p = 0.1489$). For ALT, associations were less consistent: adjusted odds of High ALT were higher at 10–19 years (OR = 1.41, 95% CI: 1.06–1.88), 30–39 years (OR = 1.69, 95% CI: 1.18–2.44) and 40–49 years (OR = 1.81, 95% CI: 1.16–2.80), but not at 20–29 years (OR = 1.34, 95% CI: 0.97–1.84) or 50 years or more (OR = 0.67, 95% CI: 0.27–1.64). Smoking for under 10 years was associated with higher odds of Borderline ALT (OR = 1.39, 95% CI: 1.04–1.86, $p = 0.0265$).

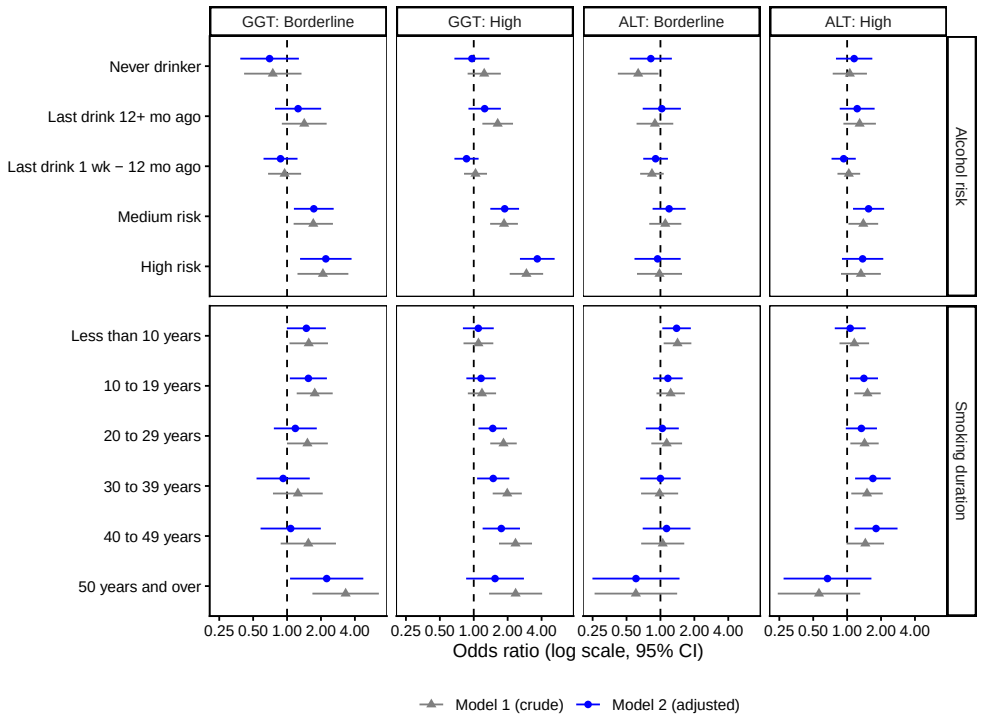


Fig 1. Odds ratios (ORs) for Borderline and High versus Low ALT and GGT across alcohol-risk and smoking-duration categories in the adult biomedical sample from the 2011–12 Australian Health Survey [30]. Separate multinomial logistic regression models were fitted for ALT and GGT. Model 1 includes alcohol risk and smoking duration simultaneously; Model 2 additionally adjusts for sex, age, waist circumference, income decile, non-school qualifications and the number of daily smokers in the household. Low-risk drinking and never having smoked daily are the respective exposure reference categories. Grey triangles represent Model 1 and blue circles represent Model 2, with horizontal lines indicating 95% confidence intervals. Odds ratios are shown on a logarithmic scale; the dashed vertical line at OR = 1 indicates no association.

Adjusted probabilities by alcohol risk. The Model 2-adjusted probability of High GGT increased from 13.3% (95% CI: 10.9–15.8%) among low-risk drinkers to 21.2% (95% CI: 16.3–26.1%) at medium risk and 33.2% (95% CI: 25.8–40.6%) at high risk, supported by the linear trend test ($p < 0.001$). The corresponding High ALT probabilities were 9.5% (95% CI: 7.7–11.3%), 13.6% (95% CI: 9.9–17.3%) and 12.6% (95% CI: 8.0–17.2%), with no graded increase (trend $p = 0.207$). The contrasting patterns were apparent on the adjusted probability scale (Figure 2).

Adjusted probabilities by smoking duration. After accounting for alcohol risk and other covariates, the adjusted probability of High GGT rose from 14.6% (95% CI: 11.6–17.5%) among those who never smoked daily to 19.5% (95% CI: 15.3–23.7%) at 20–29 years and 22.3% (95% CI: 16.6–28.1%) at 40–49 years (trend $p = 0.023$). The estimate at 50 years or more was less precise (19.0%, 95% CI: 10.8–27.3%). High ALT probabilities varied from 9.1% (95% CI: 7.0–11.1%) among never-daily-smokers to 15.0% (95% CI: 9.9–20.1%) at 40–49 years and 6.5% (95% CI: 1.1–12.0%) at 50 years or more, without a linear trend ($p = 0.980$). These associations are summarised across smoking-duration groups (Figure 3).

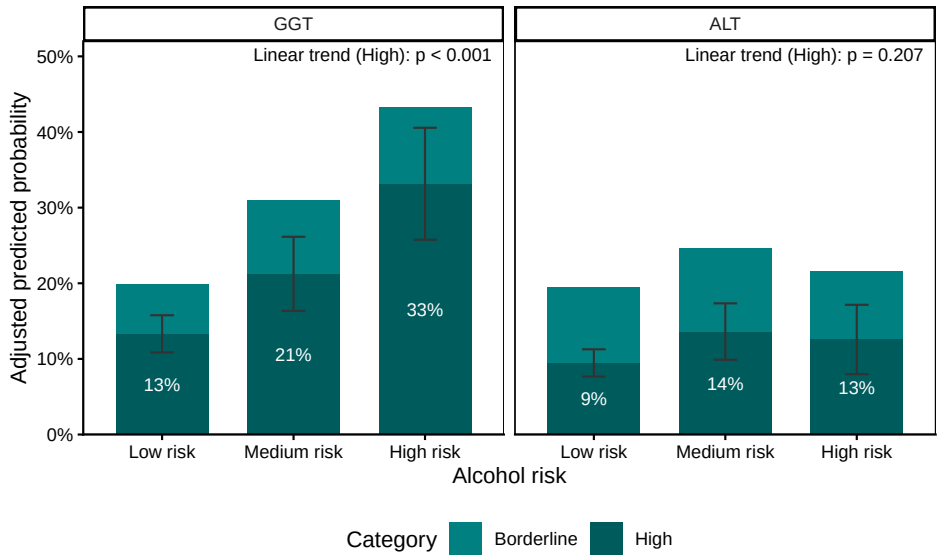


Fig 2. Adjusted predicted probabilities of Borderline and High GGT and ALT across Low, Medium and High alcohol-risk categories among current drinkers, estimated from the covariate-adjusted multinomial regression models (Model 2). The dark lower segment of each stacked bar represents High biomarker status and the lighter upper segment represents Borderline status; Low status constitutes the unplotted remainder. Percentage labels indicate the estimated probability of High status, rounded to whole percentages, with error bars showing its 95% confidence interval. Displayed p-values correspond to linear trend tests for High biomarker status across ordered alcohol-risk categories. Never and former drinkers ($n = 1720$) were retained in model estimation but excluded from these alcohol-risk probability comparisons. Model 2 adjusts for smoking duration, sex, age, waist circumference, income decile, non-school qualifications and household daily smokers.

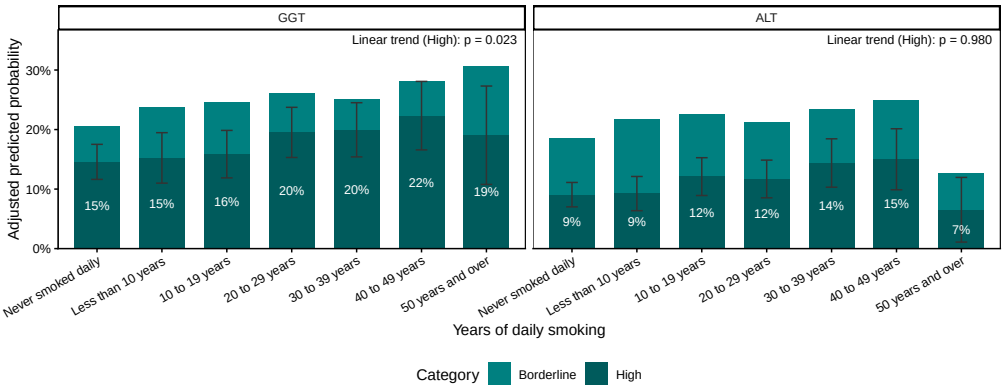


Fig 3. Adjusted predicted probabilities of Borderline and High GGT and ALT by duration of daily smoking, including participants who had never smoked daily, from the multinomial logistic regression models (Model 2). Dark lower bar segments represent High biomarker status and lighter upper segments represent Borderline status; Low status comprises the unplotted remainder. Percentage labels give the estimated probability of High status, rounded to whole percentages, with error bars representing its 95% confidence interval. Displayed p-values are linear trend tests for High biomarker status across the ordered smoking-duration categories. Model 2 additionally adjusts for alcohol risk, sex, age, waist circumference, income decile, non-school qualifications and household daily smokers.

Probabilities across all biomarker categories. The adjusted probabilities of each biomarker category (Low, Borderline or High) were compared across alcohol risk and smoking duration, accounting for both exposures and the remaining covariates (Figure 4). Alcohol risk is shown on the left and smoking duration on the right. Because Model 2 includes both alcohol risk and smoking duration, it shows how the adjusted probability of each biomarker result (Low, Borderline or High) changes across drinking risk (left) and smoking duration (right), with the other exposure and the remaining covariates held constant. For GGT, higher alcohol risk shifted participants away from a Low result (approximately 80% among low-risk drinkers to 57% among high-risk drinkers) and towards a High result (approximately 13% to 33%), whereas the probabilities of a Low (about 75–81%) or High (about 9–14%) ALT result changed little. Smoking duration produced a smaller and more gradual shift in GGT, with the probability of a High result rising from approximately 15% among those who never smoked daily to 22% among those who smoked for 40–49 years, while ALT again showed no clear direction and wide intervals at the longest duration. Taken together, once both exposures are accounted for, GGT is the biomarker that shifts with lifestyle, and alcohol risk produces the larger shift of the two.

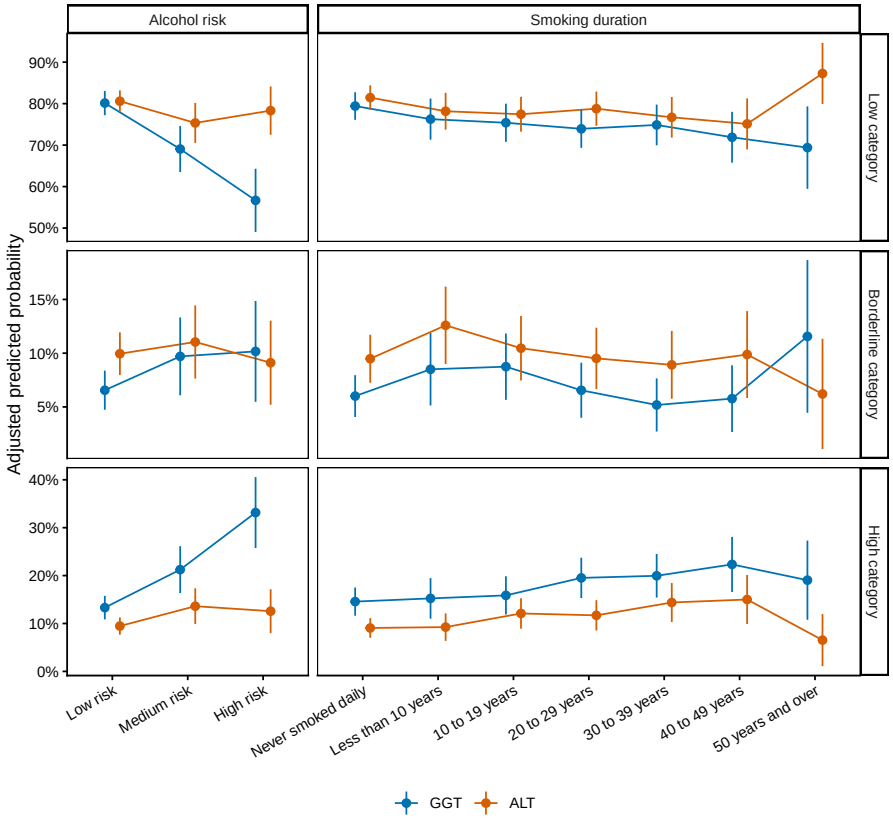


Fig 4. Model 2-adjusted predicted probabilities of Low, Borderline and High GGT (blue) and ALT (orange) across alcohol-risk categories (left column) and smoking-duration categories (right column). Points represent predicted probabilities, with vertical lines indicating 95% confidence intervals. The three rows correspond to Low, Borderline and High biomarker status and use separate probability scales. Alcohol-risk comparisons display Low-, Medium- and High-risk current drinkers; never and former drinkers remained in the fitted models but are not displayed in the alcohol-risk column. Smoking-duration comparisons include participants who had never smoked daily. Estimates account for alcohol risk, smoking duration, sex, age, waist circumference, income decile, non-school qualifications and household daily smokers.

Overall, adjusted results showed graded associations of alcohol risk and smoking duration with GGT, particularly High GGT, while associations with ALT were weaker and less consistent.

Analysis 2 Results

Physical Activity and Liver Biomarker Status. Associations between exercise-guideline status and ALT and GGT categories are presented in Table 3. In the unadjusted model (Model 1), not meeting the exercise guidelines was associated with higher relative odds of High versus Low ALT (OR = 1.30, 95% CI: 1.09–1.55, $p = 0.0036$) and High versus Low GGT (OR = 1.71, 95% CI: 1.44–2.03, $p < 0.0001$). No significant associations were observed for the Borderline categories.

After adjustment for the prespecified covariates (Model 2), the association with High ALT was attenuated and no longer statistically significant (OR = 1.11, 95% CI: 0.92–1.34, $p = 0.2727$). In contrast, not meeting the exercise guidelines remained associated with higher relative odds of High versus Low GGT (OR = 1.35, 95% CI: 1.12–1.62, $p = 0.0014$). The adjusted association with Borderline GGT was not statistically significant.

Table 3. Association between exercise-guideline status and ALT and GGT categories in the analytic sample from the 2011–12 National Health Survey, Australian Health Survey [30] ($N = 4762$), estimated using multinomial logistic regression. Model 1 includes exercise-guideline status only; Model 2 adjusts for sex, alcohol risk, smoking duration, non-school qualifications, income decile, age, and waist circumference. Odds ratios compare Exercise No with Exercise Yes, with Low ALT/GGT as the reference outcome category.

		Model 1		Model 2	
Biomarker	**Category**	**OR (95% CI)**	***p*-value**	**OR (95% CI)**	***p*-value**
ALT	Low	Ref.	–	Ref.	–
	Borderline	0.92 (0.76, 1.10)	0.3495	0.89 (0.73, 1.08)	0.2289
	High	1.30 (1.09, 1.55)	0.0036 **	1.11 (0.92, 1.34)	0.2727
GGT	Low	Ref.	–	Ref.	–
	Borderline	1.27 (0.99, 1.63)	0.0549	1.13 (0.88, 1.47)	0.3409
	High	1.71 (1.44, 2.03)	< 0.0001 ****	1.35 (1.12, 1.62)	0.0014 **

Significance codes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

Adjusted probability differences. ALT differences were small with confidence intervals crossing zero, whereas meeting the exercise guidelines was associated with approximately a +4.5-percentage-point difference in Low GGT and a −4.0-percentage-point difference in High GGT (Figure 5).

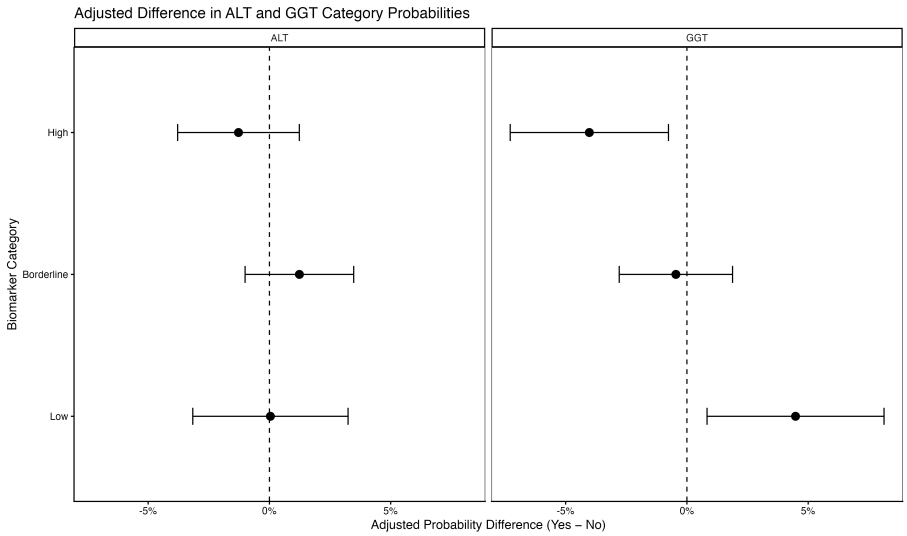


Fig 5. Adjusted Exercise Yes minus Exercise No differences in predicted ALT and GGT category probabilities in the analytic sample from the 2011–12 National Health Survey, Australian Health Survey (AHS) ($N = 4762$). Estimates are derived from multinomial logistic regression models adjusted for sex, alcohol risk, smoking duration, non-school qualifications, income decile, age, and waist circumference. Points show adjusted probability differences with 95% confidence intervals; positive values indicate higher probability among participants meeting the exercise guidelines, negative values indicate lower probability, and the vertical dashed line at zero denotes no difference.

To explore potential effect modification, adjusted probability differences were also estimated across smoking-duration and alcohol-risk subgroups.

Smoking duration. Across smoking-duration groups, the estimated exercise-associated differences were generally small and showed no clear monotonic pattern for either ALT or GGT (Figure 6). Most confidence intervals included zero, with greater uncertainty in some of the longer-duration smoking groups.

Alcohol risk. Differences across alcohol-risk groups were also generally limited for ALT (Figure 7). For GGT, the Low-risk group showed the clearest pattern, with meeting the exercise guidelines associated with a higher probability of Low GGT and a lower probability of High GGT, whereas estimates for the Medium/High-risk group were less precise and generally compatible with no difference.

Overall, the adjusted results showed little evidence of an association between exercise-guideline status and ALT category, while the clearest association was observed for GGT, with those meeting guidelines tending toward higher probability of Low GGT and lower probability of High GGT; evidence of effect modification by smoking duration or alcohol risk was comparatively weak and inconsistent.

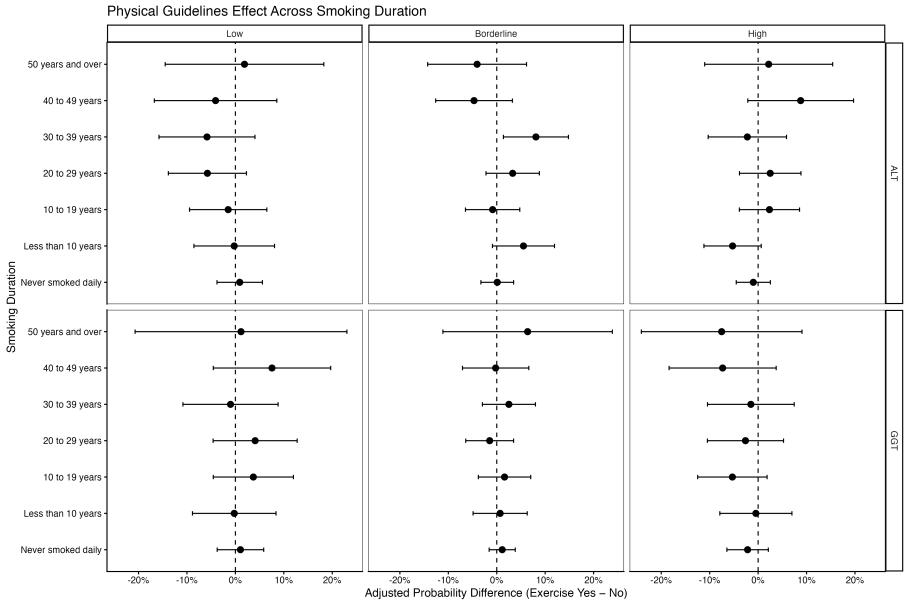


Fig 6. Adjusted Exercise Yes minus Exercise No differences in predicted ALT and GGT category probabilities across smoking-duration groups in the analytic sample from the 2011–12 National Health Survey, Australian Health Survey (AHS) ($N = 4762$). Estimates are derived from multinomial logistic regression models containing ExerciseGuidelines × SmokingDuration and ExerciseGuidelines × AlcoholRisk interactions and adjusting for sex, alcohol risk, smoking duration, non-school qualifications, income decile, age, and waist circumference. Points show subgroup-specific probability differences with 95% confidence intervals; the vertical dashed line at zero denotes no exercise-associated difference. Variation across smoking-duration groups illustrates potential effect modification by smoking duration.

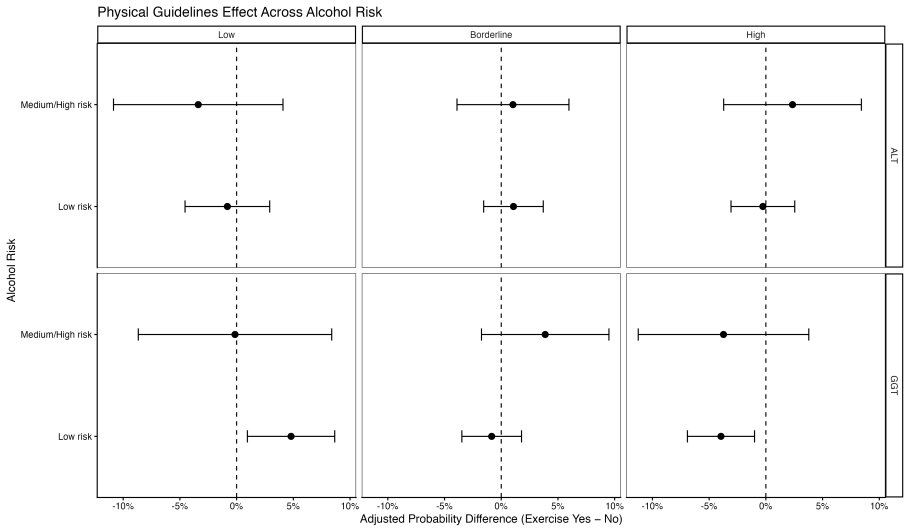


Fig 7. Adjusted Exercise Yes minus Exercise No differences in predicted ALT and GGT category probabilities across alcohol-risk groups in the analytic sample from the 2011–12 National Health Survey, Australian Health Survey (AHS) ($N = 4762$). Estimates are derived from multinomial logistic regression models containing ExerciseGuidelines × AlcoholRisk and ExerciseGuidelines × SmokingDuration interactions and adjusting for sex, alcohol risk, smoking duration, non-school qualifications, income decile, age, and waist circumference. Points show subgroup-specific probability differences with 95% confidence intervals; the vertical dashed line at zero denotes no exercise-associated difference. Variation across alcohol-risk groups illustrates potential effect modification by alcohol consumption risk.

Discussion

We examined associations of alcohol risk, smoking duration and exercise with ALT and GGT using cross-sectional data. The overall results showed dose-dependency between alcohol consumption and borderline/high GGT. High-risk alcohol consumers had 3.66 times the relative odds to have high rather than low GGT after adjustment for age, sex, waist circumference, SES, and smokers in the household. Similarly, a strong dose-dependent relationship was observed between smoking duration and high GGT. Individuals who had smoked for 20-49 years had 48-76% higher odds of high rather than low GGT after adjustment, but long smoking duration was weakly associated with higher ALT.

Not meeting exercise guidelines was associated with 35% higher relative odds of High versus Low GGT. Furthermore, within the low risk alcohol subgroup, meeting exercise guidelines was associated with a higher probability of low GGT and a lower probability of high GGT. However, among medium/high risk alcohol consumers, the exercise associated probability differences were less precise and compatible with no effect. Smoking duration on the other hand had no clear effect on the association between exercise and GGT and no other significant associations were observed with ALT after adjustment. Thus, the results partially give evidence for the hypotheses.

Principal Findings

Alcohol Risk. Several cross-sectional studies have likewise reported significant dose-dependent relationships between GGT and the amount/frequency of alcohol consumption [3, 6, 8–10, 19]. For instance, GGT was elevated in current and heavy lifetime drinkers compared to other drinking categories [10], and the comparison between different drinking frequencies (from "never" to five times per week) showed dose-dependency in average GGT levels [19]. Ethanol metabolism produces acetaldehyde, inducing oxidative stress [2, 4, 10]. This increases GSH turnover and raises GGT which leaks into the bloodstream, signifying it as a key marker of alcohol-induced hepatic inflammation [7–10]. Contrastingly though, a weak correlation between GGT and alcohol daily consumption due to the use of a presumably large and "unselected" healthy population, deemed the biomarker to be insufficiently discriminative [19]. Despite this limited conflicting evidence, GGT has mostly shown high sensitivity to alcohol-induced effects notwithstanding its lack of specificity and high susceptibility to alternative stimuli [10]. This highlights its potential use as an indicator of alcohol dependence [2, 6–8, 11].

Evidence linking ALT and alcohol consumption is mixed. Generally, heavy drinking elevates ALT due to hepatocellular damage as shown in multiple studies [2, 3, 24]. However, our results do not robustly support this association which conforms with other previous findings showing no associations both with [10] and without [6] confounder adjustment. Our conflicting results may be due to total populational alcohol consumption not being as high as other populations previously examined, or particular ethnicities being less sensitive to alcohol than others, resulting in mixed inter-individual variation [6]. In addition, the three-day recall of alcohol consumption may not be representative of an individual's habitual alcohol intake, resulting in possible misclassification of risk level. Since ALT has been demonstrated to have immediate sensitivity to alcohol withdrawal [2], misclassified moderate/high alcohol risk-level consumers may have presented normalised ALT serum levels at the time of testing.

Hence, a modest and true dose-dependent association between alcohol consumption and ALT may have been attenuated and cannot be excluded.

Smoking Duration. Furthermore, our findings of longer smoking durations, more prevalent in individuals of low SES, being dose-dependently associated with high GGT supports other cross-sectional surveys [8–10]. For example, an increase in the number of cigarettes per day was associated with an increase [8] or higher odds (in heavy smokers) [9] of high GGT. Notably, while 50+ years of smoking was associated with 135% higher odds of high rather than low GGT in the crude model, this association was no longer significant after adjustment. This could likely be confounded by age given the overlap between longest smoking duration and older age. However, one earlier finding did not support an association between elevated GGT and smoking status (presence/absence), implying that intensity or duration of smoking rather than status, has a greater effect on GGT [2].

Most previous research demonstrated no association between smoking duration and ALT [9, 10, 24], suggesting that smoking does not independently induce hepatocellular damage. However, our findings point towards a mixed pattern dependent on smoking history. Despite no significant dose-dependent relationship between smoking and ALT, those that smoked for 10 years or less had 39% higher odds of having borderline rather than low ALT, and those that smoked for 40–49 years had 81% higher odds of having high rather than low ALT, after adjustment. Smoking 50+ years was not significantly associated with ALT in either the crude or adjusted model, despite the trend across shorter durations. This is possible due to the smaller number of respondents in the longest duration group. Overall, these findings suggest that only a particularly long smoking history independently increases the risk of elevated ALT and hepatocellular damage. This may explain why the findings are in opposition to the current understanding of smoking only inducing oxidative stress and raised GGT [8–10].

Exercise Guidelines. Moreover, our results revealed that meeting exercise guidelines was associated with reduced GGT, alluding to a possible protective effect against free radical damage. Regular exercise reduces hepatic steatosis, promotes beta-oxidation [14, 26], inhibits the release of danger-associated molecular patterns and increases the release of anti-inflammatory molecules which protect against hepatocellular/mitochondrial damage [20, 21] and reduce GSH and consequently GGT requirements [16]. However, there is controversial support for the associational benefits of exercise on liver health. It was demonstrated that individuals with low physical activity were 1.4-fold more likely to have elevated GGT than highly active individuals [16]. But, sedentary time may confound and attenuate this association [26] and the duration and type of exercise, whether steady-state, interval training, or resistance, may influence this association and result in conflicting findings [20]. For example, high-intensity interval training was demonstrated to have the most significant effect on lowering liver biomarkers [21].

In the unadjusted model, not meeting exercise guidelines was associated with 30% higher relative odds of high ALT. However, this association was fully attenuated after multivariable adjustment which contrasts with the majority of literature showing a negative relationship [20–22, 27]. This may be due to the varying health context of the survey population resulting in variable ALT responses. Indeed,

previous findings in 30 male athletes found no difference in mean ALT pre-to-post aerobic training suggesting there is an upper limit at which exercise may lower ALT in healthy individuals [25]. In contrast, unhealthy individuals with higher ALT may exhibit more pronounced metabolic responses to exercise [27], thus making the results context-dependent.

Lastly, we investigated whether higher smoking durations and alcohol consumption lowered the associational significance between exercise and GGT. Though smoking duration showed no overall effect, meeting exercise guidelines exhibited a protective effect in the GGT levels of low-risk alcohol consumers, which disappeared at higher alcohol risk levels. This suggests that greater alcohol consumption may override the benefits of exercise.

Limitations and Future Directions

The cross-sectional nature of the study limits its ability to establish cause-effect relationships. Longitudinal cohort studies are required to better demonstrate the influence of lifestyle factors over time [15, 16, 24]. Also, under-reporting of alcohol/smoking and over-reporting of exercise due to desirability bias is an inherent risk to recall questionnaires, and one week of alcohol and exercise data is not representative of an individual's habitual routine. Hence, some associations may have been conservative or attenuated, particularly between alcohol consumption, exercise and ALT. Additionally, there are no consensus cut-off reference values defining abnormal ALT/GGT levels as baseline values were established using populations that included healthy and unhealthy subjects [3]. Normal ALT/GGT ranges should be redefined using populations of abstainers to improve the discriminative power of the biomarkers as indicators of alcohol use, smoking, and exercise. Future analyses should also account for the varying effect of exercise types and duration on liver biomarkers as current meta-analyses [20, 21] have not stratified for this yet. It has been suggested that longer-term exercise has greater accumulative benefits which would better inform lifestyle intervention strategies.

Methodologically, the statistical design transitioned from an ordinal to a multinomial logistic regression model following a violation of the proportional odds assumption. Furthermore, potential confounders such as age, sex, waist circumference, income, and education were systematically identified using Directed Acyclic Graphs (DAGs) to prevent confounding bias. Adjusted probability differences improved interpretability, with Holm adjustment for multiple category-specific comparisons. However, selecting a multinomial framework inherently discards the natural ordering of biomarker categories, which may reduce statistical efficiency. Exploratory generalised additive models suggested potential nonlinearity in age and waist circumference. The two analyses also yielded slightly different sample sizes, with the source of this discrepancy not fully established. Finally, complete case analysis reduced our sample sizes, resulting in wider confidence intervals and diminished power when detecting interaction effects within smaller, high exposure subgroups.

Overall, this study has shown strong associations between GGT, alcohol consumption, and smoking, weaker association between smoking duration and ALT, and has implied protective effects of exercise on liver stress. The use of GGT for the future evaluation of liver health, early diagnosis of alcohol dependence and smoking, and post-treatment monitoring has been supported through our findings. With further investigation, its utilisation can potentially allow stakeholders to better design and target prevention messaging and health-promoting policies involving exercise towards individuals with alcohol dependence and smokers, who tended to have lower SES [13, 26]. This would reduce liver-disease incidence and related healthcare costs [13].

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Author Contributions. The contributions of each author are summarised below.

Author	Contribution
Calvin Lai	Analysis 2 Methods and Results, Report Synthesis
Sameeha Shaik	Analysis 1 Methods and Results
Yukti Yadav	Analysis 1 Methods and Results
Kin Wang Yau	Analysis 2 Results, Statistical Discussion
Laila Haifa	Introduction, Nutritional Discussion, Conclusion

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